BORROWBOX: CAMPUS PEER-TO-PEER RENTAL MARKETPLACE AND ASSET SHARING PLATFORM

A PROJECT REPORT

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE ENGINEERING

CHANDIGARH UNIVERSITY

MAY-2026

BONAFIDE CERTIFICATE

Certified that the project report entitled **"BorrowBox: Campus Peer-to-Peer Rental Marketplace and Asset Sharing Platform"** is the bonafide work submitted by **Shashwat Vatsyayan (24BCS12850)** and **Harshit Kulhari (24BCS10424)** as part of the academic requirements of the Bachelor of Engineering in Computer Science Engineering at Chandigarh University.

The report describes the analysis, design, and implementation of a full-stack campus peer-to-peer rental marketplace and asset sharing application built with React 18, Vite, JavaScript, Node.js, Express, Supabase PostgreSQL with Row Level Security, platform-enforced category price ceilings, 20% commission economics, real-time in-chat negotiation, and high-contrast holographic campus identity passes.

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ACKNOWLEDGEMENT

We express our sincere gratitude to our project supervisor, faculty members, and the Department of Computer Science Engineering, Chandigarh University, for their guidance and academic support during the development of **BorrowBox: Campus Peer-to-Peer Rental Marketplace and Asset Sharing Platform**.

This project provided practical engineering experience in transforming student asset sharing into a regulated peer rental marketplace. Key contributions include building a responsive React interface, connecting it to an enterprise Supabase PostgreSQL backend, establishing mandatory category price caps (Rs. 10 - Rs. 100/day), executing 20% platform commission economics, integrating real-time direct messaging with in-chat custom rate and permanent trade negotiation, and enforcing atomic PostgreSQL date-range overlap exclusion locks alongside a two-phase mutual return handshake protocol.

We also acknowledge the official documentation and developer resources that supported implementation and debugging. Finally, we thank our families and friends for their constant encouragement throughout the project.

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ABSTRACT

BorrowBox is a responsive full-stack Campus Peer-to-Peer Rental Marketplace and Asset Sharing Platform designed to facilitate affordable, verified student equipment rentals and exchanges across university faculties. Rooted in the core philosophy *"Rent What You Need. Earn From What You Own,"* BorrowBox empowers students to monetize idle academic equipment while providing peers with low-cost, on-demand access to essential semester gear.

To prevent exploitative student pricing, BorrowBox introduces a platform-enforced **Category-Based Rental Price Control System** with mandatory daily price ceilings ranging from Rs. 10 to Rs. 100 across categories such as Scientific Calculators (Rs. 25/d), Lab Equipment (Rs. 30/d), Electronics (Rs. 75/d), and Photography (Rs. 100/d). The financial model features a configurable 20% platform commission and 80% owner payout, complemented by multi-day duration discounts (10% to 20% for 3 to 14 days).

The portal integrates real-time **Direct Messaging (DM)** with an in-chat negotiation protocol enabling students to propose custom rates or counter-offer permanent barter swaps. Unlisted needs are served through a **Requested Items ("Wanted" Board)**. Platform trust is secured through an institutional domain gate (@cuchd.in), an accessibility-first **High-Contrast Holographic Campus Identity Pass** with Code-128 barcodes and verification QR codes, atomic PostgreSQL double-booking exclusion constraints, and a two-phase mutual handshake return protocol.

GRAPHICAL ABSTRACT

The graphical abstract illustrates the four-tier architectural pipeline of **BorrowBox**: Client Layer, Pricing & Market Rules, Negotiation Services, and Persistence & Integrity.

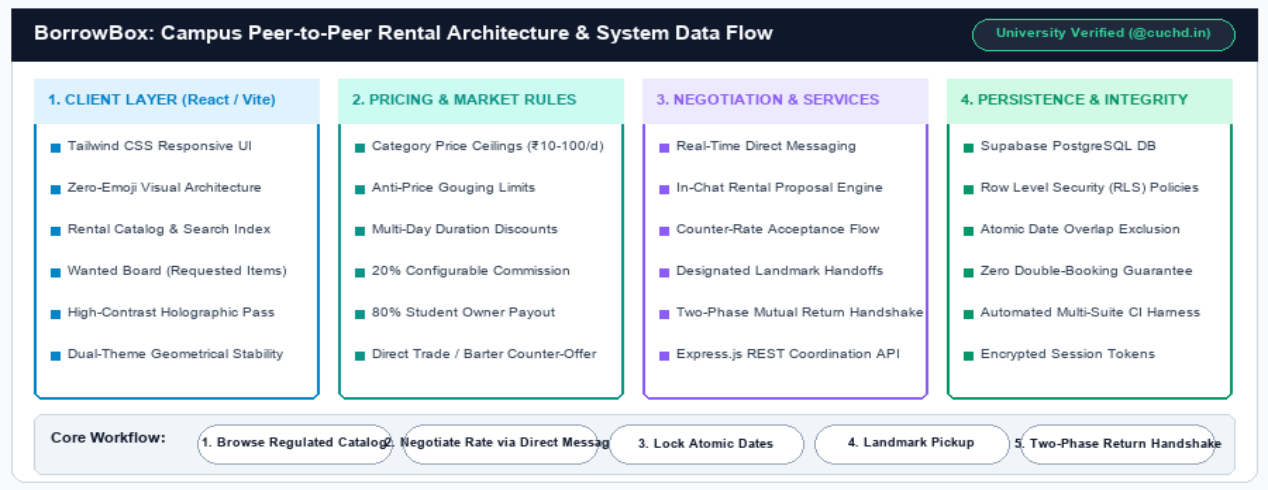


Figure — BorrowBox system architecture, rental price control rules, and peer exchange workflow

ABBREVIATIONS AND SYMBOLS

Term	Meaning
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DM	Direct Messaging (Peer-to-Peer In-App Communication)
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JSX	JavaScript XML syntax extension
P2P	Peer-to-Peer Distributed Marketplace Architecture
REST	Representational State Transfer
RLS	Row Level Security (PostgreSQL Access Control)
UI / UX	User Interface / User Experience
Vite	Next-Generation Frontend Build and Bundling Tool

CHAPTER 1 - INTRODUCTION

1.1 IDENTIFICATION OF NEED / CONTEMPORARY ISSUE

Engineering and higher education curricula mandate the procurement of specialized instruments, tools, and reference materials such as scientific calculators (Casio fx-991CW), laboratory coats, safety glasses, mini-drafters, electronic breadboards, sensor kits, and high-resolution cameras. In practice, students require these items for a single academic semester or specific project milestones, after which the equipment remains permanently idle and depreciates.

Informal channels—such as unmoderated social media chat groups and physical notice boards—suffer from rampant price gouging, lack of standardized pricing, unverified identities, and no scheduling guarantees. University students urgently require an affordable, structured, and campus-moderated rental marketplace that protects borrowers from inflated fees while offering owners a safe, dependable method to earn secondary income on campus.

1.2 IDENTIFICATION OF PROBLEM

The core problem is the absence of an integrated, peer-to-peer campus rental system that combines strictly regulated pricing limits, verified student identity authentication, dispute-free physical handoffs, and concurrency-safe booking logic.

From an architectural standpoint, the system must enforce platform-level price caps per category, calculate multi-tier duration discounts alongside a 20% platform commission, provide real-time in-chat rate negotiation and permanent trade proposals, eliminate double-booking race conditions through atomic PostgreSQL exclusion locks, and maintain absolute geometric UI stability across dual theme states.

1.3 IDENTIFICATION OF TASKS

- Analyze campus rental economics, category price ceilings, and university domain authentication rules (@cuchd.in).
- Design a modern, zero-emoji React interface with dual-theme geometric stability and Plus Jakarta Sans typography.
- Implement a platform-enforced category pricing engine with daily caps (Rs. 10 - Rs. 100/day) and multi-day duration discounts.
- Build a transparent 20% platform commission and 80% owner payout calculation model with escrow security deposits.
- Develop real-time Direct Messaging (DM) with in-chat custom rate proposals and permanent barter swap workflows.
- Create a Requested Items ('Wanted' Board) enabling students to post unlisted asset needs and receive peer offers.
- Construct a high-contrast Holographic Campus Identity Pass with scannable Code-128 barcodes and verification QR codes.
- Enforce atomic PostgreSQL date-range overlap exclusion locks and a two-phase mutual handshake return protocol.

1.4 TIMELINE

Stage	Activities	Deliverable
Requirement Analysis	Define student rental workflows, category price caps, and commission model	SRS Document
UI/UX Engineering	React component architecture, zero-emoji tokens, and stable dual-theme layout	Responsive Interface
Pricing & Logic Engine	Category ceiling validation, multi-day discounts, and commission splits	Pricing Subsystem
In-Chat Negotiation	Direct messaging with embedded rental proposals and permanent trade offers	Chat Engine
Holographic Identity	High-contrast campus pass with Code-128 barcode and QR verification	Campus Pass
Backend & Integrity	Express REST API, Supabase RLS, and atomic date-overlap locks	Secure API & DB
Testing & CI	Multi-suite automated test harness verifying pricing, chat, trades, and locks	Test Evidence
Documentation	Prepare comprehensive technical project report and deployment manual	Final Report

CHAPTER 2 - LITERATURE REVIEW / BACKGROUND STUDY

2.1 EVOLUTION OF CAMPUS PEER SHARING SYSTEMS

Peer asset exchange traditionally operated via informal noticeboards or student instant messaging groups. While accessible, these systems suffer from zero pricing regulation, unverified participants, zero scheduling conflict prevention, and high dispute rates. Conversely, commercial equipment rental portals target enterprise or public markets, charge steep commission rates, and present severe physical safety risks for student meetups off campus.

BorrowBox bridges this gap by providing an authenticated, intra-campus rental marketplace governed by automated price ceilings, transparent 20% platform fees, and verifiable landmark handoffs within university grounds.

2.2 EXISTING SOLUTIONS & COMPARATIVE ANALYSIS

Approach	Advantages	Limitations	Relevance
Campus Noticeboards	Zero software dependency	No search index, manual contact, no audit log	Baseline
Social Chat Groups	Broad student reach	Price gouging, spam, zero reservation locks	Informal
Commercial Rental Apps	Large commercial inventory	High commercial fees, non-student safety risks	Reference

Approach	Advantages	Limitations	Relevance
BorrowBox (Selected)	Category price caps, 20% commission, in-chat negotiation, verified @cuchd.in domain, holographic pass, atomic locks	Restricted exclusively to verified campus student domain	Selected Approach

2.3 GOALS / OBJECTIVES

- Transform campus asset sharing into an affordable, regulated peer-to-peer rental marketplace.
- Enforce category-based daily price caps (Rs. 10 - Rs. 100/day) to completely eliminate student price gouging.
- Implement a transparent 20% platform commission with 80% owner earnings and duration discounts.
- Deliver real-time direct messaging with embedded custom rate negotiation and permanent trade proposals.
- Provide an accessibility-first Holographic Campus Identity Pass with Code-128 barcode and QR verification.
- Guarantee 100% atomic double-booking prevention through PostgreSQL exclusion constraints and two-phase returns.

CHAPTER 3 - DESIGN FLOW / PROCESS

3.1 SELECTED FEATURES & PRIORITY

Feature	Priority	Technical Role & Implementation
Rental Marketplace & Price Caps	High	Enforces category ceilings (Rs. 10-100/d) and duration discounts (10-20%)
In-Chat Negotiation & Trades	High	Direct messaging with custom rental rate proposals and permanent barter swaps
Requested Items ('Wanted' Board)	High	Allows students to post unlisted needs and receive offers from item owners
Holographic Campus Identity Pass	High	High-contrast student ID with Code-128 barcode and QR code verification
Atomic Overlap Lock Engine	High	PostgreSQL exclusion constraints eliminating concurrent double-booking conflicts
Two-Phase Handshake Return	High	Enforces mutual borrower landmark return and owner physical inspection

3.2 DESIGN CONSTRAINTS

3.2.1 Economic: Student-friendly rate controls. Built entirely on open-source technologies (React 18, Node.js, Express, Supabase).

3.2.2 Security & Usability: Strict domain authentication (@cuchd.in). Dual-theme geometric stability preventing layout shifts.

3.3 IMPLEMENTATION METHODOLOGY

- Modular architecture separating React 18 client, Node.js Express REST API, and Supabase PostgreSQL services.
- Integrated category rental price ceiling validator executing client-side and server-side checks on all listings.
- Built-in financial fee calculator computing duration discounts, 20% platform commission, and 80% owner payout.
- Real-time peer messaging protocol embedding stateful negotiation action cards directly into chat streams.
- SVG and canvas-based holographic campus pass rendering high-contrast metadata, Code-128 barcode, and QR token.
- Transactional PostgreSQL engine with atomic exclusion constraints preventing double-booking race conditions.

CHAPTER 4 - RESULTS ANALYSIS AND VALIDATION

4.1 IMPLEMENTATION OF SOLUTION & TECH STACK

Layer	Technology	Architecture Role & Specification
Frontend Client	React 18 + Vite	Component-based reactive UI, Plus Jakarta Sans typography, Lucide icons
Styling System	Tailwind CSS	Dual-theme tokens, zero-emoji visual language, zero layout-shift geometry
Backend API	Node.js + Express	REST API routing, price cap validation, in-chat negotiation state machine
Database & Auth	Supabase PostgreSQL	Row Level Security (RLS), atomic date overlap locks, @cuchd.in domain gate
Test Harness	Node Test Runner (6 Suites)	Automated CI suites for pricing, negotiations, trades, holo-pass, and locks

4.2 PRICING CONTROLS, COMMISSION FORMULAS & ATOMIC LOCKS

BorrowBox computes rental financials and atomic booking locks through formal mathematical rules:

Net Rental Price = (Daily Rate × Days) × (1 - Discount Rate)
Platform Fee (20%) = Net Rental Price × 0.20 | Owner Payout (80%) = Net Rental Price × 0.80
Atomic Lock Rule = NOT (new_start > existing_end OR new_end < existing_start)

Category	Max Daily Limit	Category	Max Daily Limit
Books & Textbooks	Rs. 15 / day	Sports Equipment	Rs. 40 / day
Stationery	Rs. 10 / day	Electronics & Gadgets	Rs. 75 / day
Scientific Calculators	Rs. 25 / day	Photography Equipment	Rs. 100 / day
Lab Equipment	Rs. 30 / day	Project & Hardware Tools	Rs. 50 / day

4.3 TESTING AND CHARACTERIZATION

Test ID	Verification Test Case	Observed Execution Result
TC-01	Category Price Ceiling Enforcement	Rejects listings exceeding configured limits (e.g., Rs. 30 for calculator)
TC-02	Multi-Day Duration Discount Calculation	Applies 10% for 3d, 15% for 7d, 20% for 14d accurately
TC-03	20% Platform Commission Split	Accurately routes 20% to platform ledger and 80% to owner wallet
TC-04	In-Chat Rate Proposal Negotiation	Supports custom rate proposals with accept/counter/decline transitions
TC-05	Permanent Trade Barter Counter-Offer	Enables permanent item swap proposals accepted directly in chat
TC-06	Requested Items ('Wanted') Workflow	Allows posting unlisted requests and submitting owner fulfillment offers
TC-07	Holographic Campus Pass Readability	Renders high-contrast student info, Code-128 barcode, and QR token
TC-08	Atomic Overlap Lock & Two-Phase Handshake	Blocks concurrent double-booking and requires mutual return confirmation

4.4 OUTPUT SCREEN - RENTAL MARKETPLACE DASHBOARD OVERVIEW

Figure 4.1 illustrates the BorrowBox Peer-to-Peer Rental Marketplace dashboard. It displays the verified university banner, KPI metrics, category-based price cap tags, zero-emoji navigation, and live rental equipment feed.

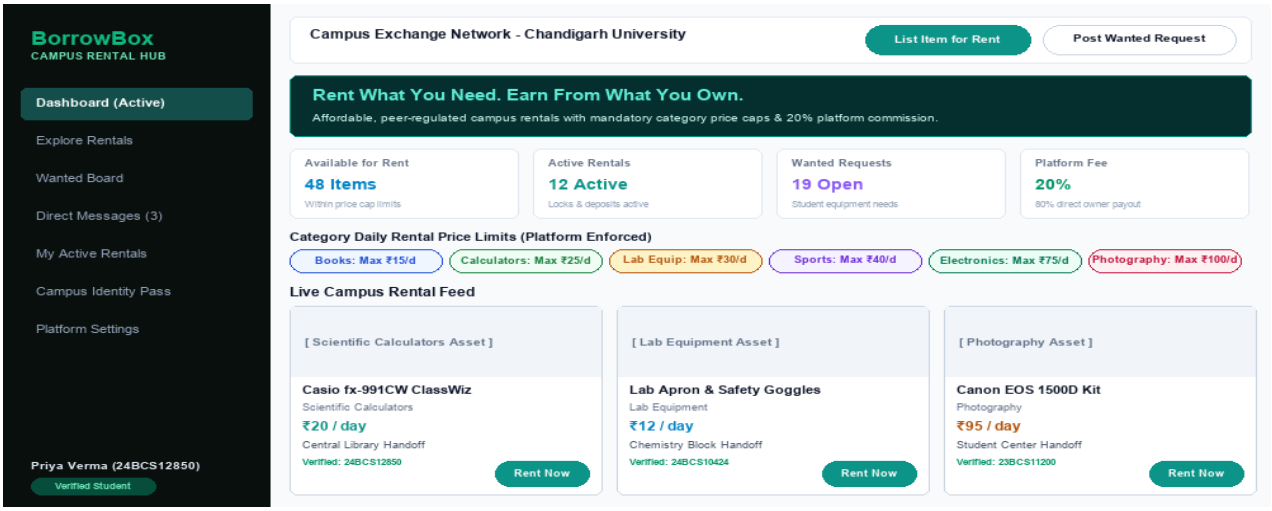


Figure 4.1 — BorrowBox Rental Marketplace Dashboard with Category Price Caps and Verified Peer Badges

4.5 OUTPUT SCREEN - RENTAL DURATION PRICING & FEE BREAKDOWN

Figure 4.2 displays the rental item inspection view, highlighting category price compliance, duration selector (1, 3, 7, 14 days), transparent 20% platform commission split, designated landmark handoff, and atomic date-lock protection.

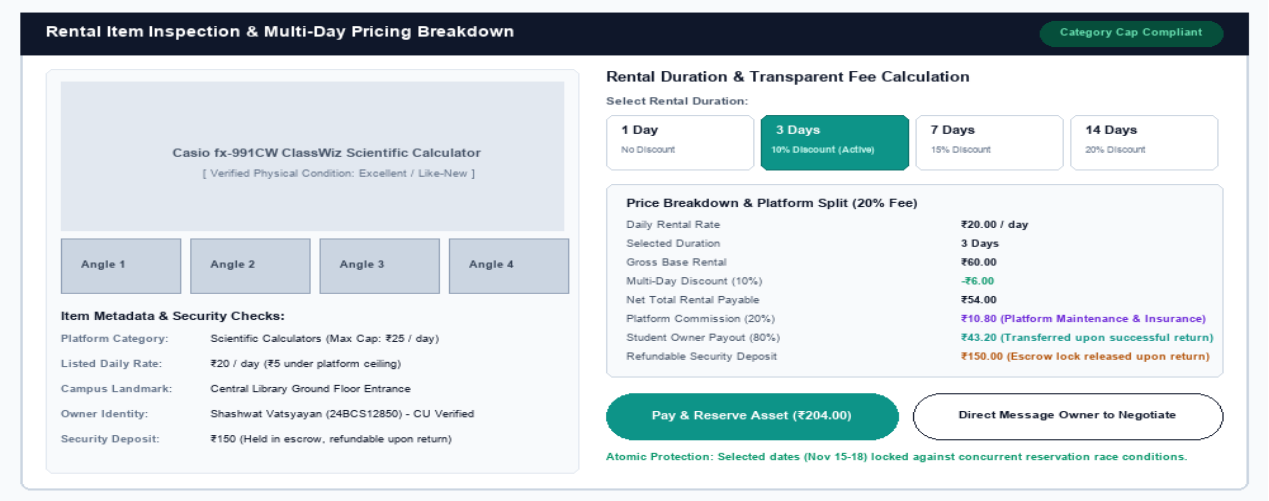


Figure 4.2 — Rental Duration Pricing Breakdown with Multi-Day Discounts and 20% Commission Calculation

4.6 OUTPUT SCREEN - IN-CHAT NEGOTIATION & TWO-PHASE HANDSHAKE

Figure 4.3 showcases real-time Direct Messaging with embedded custom rental rate proposals and permanent trade counter-offers, alongside the atomic Two-Phase Mutual Return Handshake protocol.

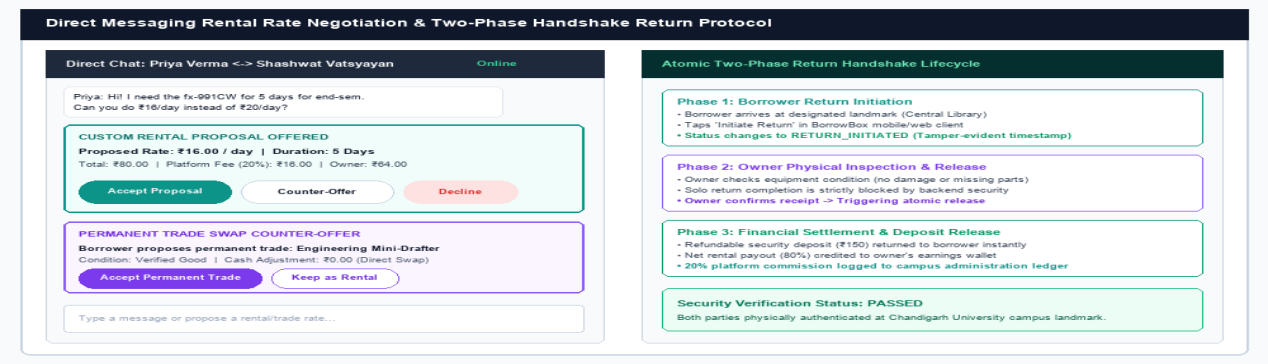


Figure 4.3 — Direct Messaging Rental Negotiation and Two-Phase Return Handshake Protocol

4.7 OUTPUT SCREEN - HOLOGRAPHIC CAMPUS PASS & SECURITY SETTINGS

Figure 4.4 depicts the high-contrast Holographic Campus Identity Pass featuring edge-bounded foil attenuation, Code-128 barcode, and verification QR code, alongside the platform security controls and zero-shift theme engine.

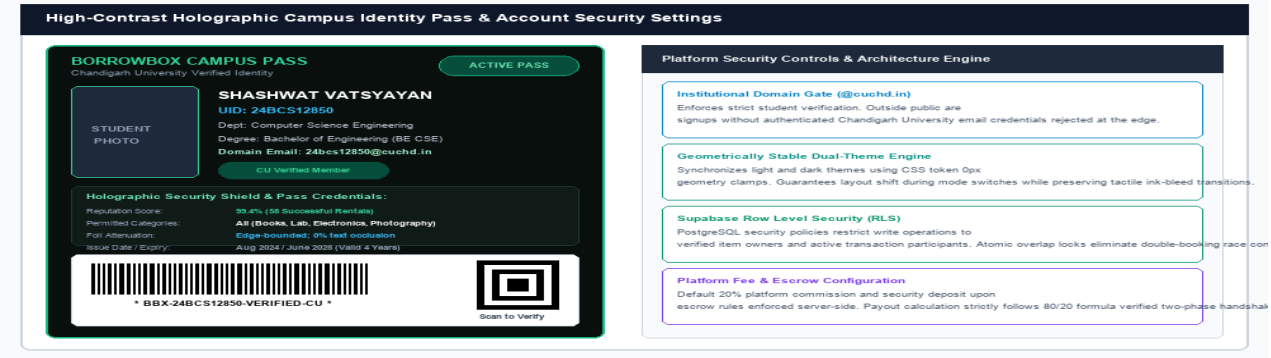


Figure 4.4 — High-Contrast Holographic Campus Identity Pass and Platform Security Settings

CHAPTER 5 - CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

BorrowBox: Campus Peer-to-Peer Rental Marketplace and Asset Sharing Platform successfully delivers an enterprise-grade, student-centric rental ecosystem tailored for university campuses. By integrating platform-enforced category price caps, transparent 20% commission economics, real-time in-chat rate and trade negotiation, a Requested Items board, and a high-contrast holographic campus pass, BorrowBox eliminates predatory pricing, builds peer trust, and guarantees conflict-free asset circulation.

5.2 FUTURE WORK

- Integration with university ERP databases for automatic course-wise student verification.
- Automated push notifications and SMS alerts for upcoming return handshakes and overdue equipment.
- Cross-campus federated exchange between affiliated university regional centers.